

Meta-Diffusion: work allocation for the next stage

Three parallel packages, mapped to *Immediate Next Steps* (9 September)

10 September 2026

Repository	CeciliaCaimi/Neural-SDE-Meta-Learning-, branch meta-diffusion-clean
Baseline commit	532361d
Per-package brief	handoff/<Name>/README.md
Fixed values	docs/PROTOCOL_CARD.md

1 Correspondence with the source note

Every item below is keyed to a numbered section of *Meta-Diffusion: Immediate Next Steps*, dated 9 September. Two pieces of work are **additions** that the note does not ask for; both are marked as such here rather than folded in silently, and each is justified where it appears.

Note	Work item	Owner	What it produces
§1	Generation-level task specificity (E12)	Jinjian	Four coordinates, one initial noise; four measurements and a paired visual grid
—	Post-hoc controls (E12a), addition	Jinjian	The mean-coordinate substitution on the two full-width checkpoints, which no log on disk contains, plus the coordinate-loss profile
§2	Centred coordinate (E13, part one)	Jing Peng	$z_y = \bar{z} + \delta z_y$, the shared offset absorbed into the base predictor
§3	Four-way panel against step (E13, part two)	Jing Peng	$\mathcal{L}(z_y)$, $\mathcal{L}(\bar{z})$, $\mathcal{L}(z_{\text{shuffled}})$, $\mathcal{L}(0)$; Δ_{task} ; basis usage and coordinate spread
§3	Calibration of Δ_{task} on stage 1, addition	Yu Cao	The same panel in the arm where the mechanism demonstrably works, plus Δ_{task} against task distance
§4	Deployment sweeps	Jinjian	The harness, unpinned and extended; the sweep itself runs at integration
§5	Critical novelty test (E15)	Jing Peng	Generic latent conditioning against the reverse-dynamics basis, same transport map
§6	DomainNet gate	—	Not started, and not startable until both conditions hold
§7	Excluded fixes	all three	A constraint, not a task. See section 8
§8	Output for the next meeting	mapped	See section 7

Why the two additions. The note’s §1 asks for the generation test *before* changing the model. E12a runs first and costs ten minutes: the mean-coordinate substitution has never been applied to the two 128-channel checkpoints, because the control was added to the code after every CIFAR run had already finished. Its result may reframe §1 before any generation code is written.

The second addition is more consequential. The quantity §3 makes decisive, Δ_{task} , has never been measured anywhere it is known to be positive. Stage 1 — two-dimensional Gaussian mixtures — computes the correct coordinate against zero and against a shuffled coordinate, and has never computed the mean-coordinate substitution. A zero on CIFAR therefore has no calibration: nobody knows whether this diagnostic can produce a large positive number at all, in this codebase, on any domain. Establishing that costs three hours of laptop CPU.

Work completed before the split. Four items had to be settled in advance because all three packages depend on them: the frozen three-corruption task family `artifacts/cifar100_domainshift_c3.json` (the split builder previously existed for no runner to call); a `model_cls` seam in `training/loop.py` so the comparison arm needs no edit to the loop both arms share; `--domainshift-path` on the runners; and `CIFAR100_ROOT` so machines that keep the dataset elsewhere need no code change. All are in the baseline commit.

2 Allocation, and why

	Machine	Branch	Why this machine
Jinjian	RTX 5080, local	jinjian	Holds the only copy of the three trained checkpoints (430 MB each, not in git) and the CIFAR-100 data. Sampling 20,480 images needs a CUDA device, but only for minutes
Jing Peng	AWS GPU server	jing-peng	5.5 GPU-hours of training, and the only machine that can run several of them at once
Yu Cao	Mac, CPU only	yu-cao	Stage 1 is synthetic, so no dataset and no transfer; measured at 54 iterations per second on CPU, which is 80,000 steps in 24 minutes

The three packages are independent by construction. Everything shared was fixed before the split; every judgement is deferred to integration; and each file in the repository has exactly one owner. Nobody needs to wait on, or ask, anyone else.

3 Rules that apply to all three

- **One owner per file.** The table is in `docs/PROTOCOL_CARD.md`. If a file is not yours, do not edit it, even to fix an obvious bug — record the bug in your results file instead.
- **Deliver numbers, not conclusions.** Whether the method passes is decided at integration, against criteria fixed before anyone sees a result.
- **Report absolute paired differences with 95 % intervals, and state the step count.** Do not headline `task_specific_frac`: its denominator decays towards zero and changes sign, so the ratio is unstable exactly where the reading matters.
- **Read nothing before the curve has levelled.** Quote the mean over the final fifth of training and the slope over the last 10,000 steps.
- **Pair everything.** All controls share one noising draw. Between-episode variance is roughly sixteen times the effect; the paired difference has a standard deviation sixty-five times smaller than the raw loss, so a raw loss curve cannot show this effect and will mislead anyone who reads it.
- **Results go in `handoff/<Name>/results/`, which is tracked.** Not `artifacts/`, which ignores `.txt`, `.png` and `checkpoints/`, so files put there vanish without an error.

4 Jinjian — Measurement

Answers §1; prepares §4. About 5 hours of development, 1.5 hours of GPU. No training.

1. **Post-hoc controls** (**addition**, 10 minutes of GPU). Rebuild each model from the checkpoint’s own stored configuration, apply the moving-average weights, and evaluate the four coordinates on 24 validation episodes with four noise draws each, reporting per-episode paired differences. Then trace the coordinate-loss profile, $\mathcal{L}(\bar{z} + s(z_y - \bar{z}))$ against $\mathcal{L}(\bar{z} + s(z_j - \bar{z}))$ for $s \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2\}$. Flat along the correct ray and rising along wrong rays would mean the basis expresses distance from a centroid and nothing else.
2. **Generation-level task specificity** (§1). Twenty validation classes, four coordinates — own, mean, shuffled, zero — and 256 samples each, by DDIM at $\eta = 0$ over 50 steps, with *identical initial noise across all four*. Four measurements: transformation consistency, semantic-class consistency, distributional distance, and task-identification accuracy against a chance baseline of 5%. Plus a paired visual grid, three episodes chosen before looking at the numbers.
3. **Sweep harness** (§4). Unpin the checkpoint in `scripts/phase_d_sweeps.sh`, and add the one statistic the table does not currently print: the value of K_T at which target-only overtakes transport. Coordinate space already places that crossover between 5 and 20.

Departure from the note, to be stated in the results file. §1 asks for the kernel Inception distance. Inception is a dependency this repository does not carry, and with 115 real images per class the estimate would be dominated by its own variance. Section 9 of `ALGORITHM.md` already prescribes sliced Wasserstein or energy MMD for this case. The standard number can be added at integration on pooled samples.

Deliverables. `E12a_controls.txt`, `E12_generation.txt`, `E14_harness_check.txt`, `coord_profile.png`, `gen_grid.png`.

5 Jing Peng — Training

Answers §2, §3 and §5. About 5.5 hours of development, 5.5 hours of GPU; the runs are independent and parallelise across instances.

1. **Centred coordinate** (§2). Write

$$\hat{\epsilon}_y(x, t) = \underbrace{[\hat{\epsilon}_0(x, t) + B(x, t) \bar{z}]}_{\text{shared predictor}} + B(x, t) \delta z_y,$$

as a buffer holding a running mean, subtracted inside the single place every coordinate already passes through. Roughly fifteen lines; no new network. Three consequences must be decided rather than discovered: the null control moves from $z = 0$ to $z = \bar{z}$, so `loss_zero` leaves the centred panel; the bit-for-bit identity assertion in `tests/test_score_identity.py` is parameterised on the flag rather than deleted; and the drifting buffer is excluded from the moving-average shadow, which otherwise shadows every floating-point entry of the state dictionary and would leave the diagnostics reading a centre training never used.

2. **The decisive diagnostic** (§3). Log all four losses with per-episode paired differences, alongside r_{basis} , coordinate spread and the norm of the running centre. Raise the diagnostic sample from 8 episodes to 24 and from one noise draw to four. The field names `delta_task` and `z_center_norm` are an interface: another package’s plotting script reads them.
3. **Four runs.** The 32-channel pair first, about an hour together; then the 128-channel pair, about two and a half hours. The centring-*off* runs are not optional and not merely a control

— no run on disk has ever logged the mean-coordinate panel, and two of the four deliverables §8 asks for are that curve.

4. **Critical novelty test** (§5). Generic latent conditioning entering the U-Net through the feature-wise modulation path that time already uses, with the same transport map, matched on all seven conditions the note lists. Because every downstream component reaches the model through one method, the substitution is a drop-in and the matching holds structurally rather than by bookkeeping. Report both parameter counts rather than asserting the budgets are equal.

Deliverables. `E13_panel.txt`, `E15_film_panel.txt`, and copies of the `*.log.jsonl` training logs — easy to forget, since the originals are written where git ignores them, and they are the primary evidence. Checkpoints are 430 MB and stay on the server.

6 Yu Cao — Positive control

Calibrates §3. About 3 hours of development, 3 hours of CPU. No GPU, no dataset, no file from anyone else’s machine.

1. **The four-way panel on stage 1.** Stage 1 already reports basis usage, the correct coordinate against zero, a shuffled coordinate, and coordinate spread. Add the mean-coordinate substitution, per-task paired intervals, and the coordinate-loss profile. On images the profile is expected to be flat along the correct ray; here a clear minimum at $s = 1$ is expected, and the contrast between the two is the point.
2. **Task distance.** The stage-1 family has a dial images do not: `--family related --perturb p` interpolates from every task identical to mutually unrelated. The repository states a falsifiable claim — *the method needs a task family whose members lie farther apart than the shared backbone can absorb* — which predicts that Δ_{task} rises monotonically with p . It has never been tested. Five points, two hours of CPU, and the training script already exists.
3. **The shared plotting tool,** used by everyone at integration. Two panels: Δ_{task} against step with an interval band, visually dominant; below it basis usage, coordinate spread and the centre norm. It must tolerate every new field being absent, since no log currently on disk has any of them, and it must shade the final fifth of training, which is the only window a number may be read from.

Deliverables. `stage1_panel.txt`, `stage1_relatedness.txt`, `stage1_profile.png`, `delta_task_vs_distance.png`.

7 Output for the next meeting

The four items §8 requests, and who produces each.

§8	Requested	From
1	Generation comparison across $z_y, \bar{z}, z_{\text{shuffled}}, z = 0$	Jinjian
2	One centred-coordinate run showing the four losses against training step	Jing Peng
3	Basis-usage and coordinate-spread curves	Jing Peng, plotted with Yu Cao’s tool
4	A clear conclusion on whether task specificity survives at convergence	integration

Item 4 is deliberately nobody’s individual deliverable. The three packages deliver numbers; the two gates — does generation differ across coordinates, and is $\Delta_{\text{task}} > 0$ at convergence —

are decided afterwards, against thresholds written down before any result was seen. Also at integration: the final deployment sweep on whichever checkpoint passes, and Jinjian’s generation instruments applied to Jing Peng’s comparison arm.

Alongside item 4, the stage-1 calibration is what makes it readable. A zero on images means one thing if the same measurement is large and positive on Gaussian mixtures, and something quite different if it is near zero there too.

8 Out of scope

§7 rules out, as the primary fix: a larger k ; another backbone size; a more complex transport network; confidence gates; separate source and target encoders; probabilistic latent machinery; and healthcare- or intervention-specific components. Nothing proposed above adds a network.

The first two are not merely discouraged — they are already done, and negative. Coordinate dimension $k \in \{16, 32, 64\}$ was swept at matched steps: the early peak rises with k and all three decay to between 0.0004 and 0.0007, which is the signature of a backbone absorbing the task difference rather than of a coordinate short of capacity. Backbone width was swept at fixed k : apparent gain rises 725-fold as the backbone shrinks, and the mean-coordinate substitution then showed that none of even that was task-specific. Both sweeps were read as successes at the time and both readings were wrong. Repeating either would cost a day to reproduce a number already in `artifacts/`.

§6 keeps DomainNet closed until two conditions hold: $\Delta_{\text{task}} > 0$ at convergence, and generation from z_y measurably different from generation from $\bar{z}$. Neither has been tested yet, and neither becomes more likely with a larger dataset.

One open question

The allocation assumes the AWS instance carries a GPU, and benefits from more than one. If it does not, Jing Peng’s package has nowhere to run and the split must be revised before anyone starts.